MIL-S-22135C(OS)
20 October 1972
Superseding
MIL-S-22135B(OS)
15 June 1967

MILITARY SPECIFICATION

SWITCH, FIRING, MARK 9 MOD 0

(For Exploder Mechanism Mark 19 Series)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification states the requirements for the procurement of Firing Switch Mk 9 Mod 0, LD 161902, an inertia-operated electric switch with normally open contacts which functions on target impact to complete a firing circuit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Stayed (Including Stay Material)

Military

MIL-P-116	Preservation, Methods of
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for

FSC 1355

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

LD 161902	Firing Switch Mark 9 Mod 0 and all parts and subassemblies listed thereon
Dwg 771396	Firing Switch Inner Assembly
Dwg 1576934	Switch Base Assembly

PUBLICATIONS

Naval Ordnance Systems Command

OR-11	Development of Packaging, Handling, Storage and Transportation Systems for Weapons

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Firing Switch Mk 9 Mod 0 is an inertia-operated, ball-type electric switch with normally open contacts. It consists of a ball weight assembly and terminal assembly encased in a hermetically sealed metallic shell. The shell is completely encased in a plastic housing. Two contact pins are connected internally with the ball and the shell and plug into an electrical receptacle at the base of the cavity where the switch is installed.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the firing switch shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's

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production methods are capable of producing switches that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of Firing Switch Mk 9 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 161902.

3.2.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for Firing Switch Mk 9 Mod 0 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 161902.

3.2.4 Moisture, fungus, and corrosion. Firing Switch Mk 9 Mod 0 shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications in LD 161902.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 161902.

3.2.6 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Performance requirements and product characteristics. The firing switch shall meet the following performance requirements and product characteristics:

3.3.1 Airtightness. The firing switch inner assembly, Dwg 771396, shall evidence no leakage of air when the external air pressure is reduced to 15 inches of mercury.

3.3.2 Sensitivity and contact resistance

(a) When the firing switch is accelerated in such a direction that the ball is urged along the main axis toward the nose cap, the switch shall not close with an applied acceleration up to 4.0 gravity units and shall reach a state of steady closure before the applied acceleration reaches 7.0 gravity units.

(b) When the firing switch is accelerated in such a direction that the ball is urged along the main axis toward the base, the switch shall not close with an applied acceleration up to 2.5 gravity units and shall reach a state of steady closure before the applied acceleration reaches 4.0 gravity units.

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(c) When the switch axis is vertical and the switch is in a base-down position, the switch shall not close with an applied acceleration up to 2.0 gravity units and shall reach a state of steady closure before the applied acceleration reaches 4.5 gravity units.

(d) The contact resistance at steady closure shall be 0.5 ohms or less. Steady closure shall be considered obtained when a straight line (with possible hash due to circuitry) is observed on the oscilloscope.

3.3.3 Dielectric requirement. When the firing switch is in the normally open condition, the insulation resistance, as measured between the terminals of the switch, shall be not less than 10 megohms under an applied potential of 500 volts direct current (dc).

3.3.4 Operating life. The firing switch shall have an operating life of not less than 100 complete cycles as defined in 4.4.5.1. At the conclusion of the operating life test, the firing switch shall meet the performance requirements and product characteristics of this specification.

3.4 Environmental requirements

3.4.1 Vibration

3.4.1.1 Vibration; transportation, storage, and handling. The firing switch shall be capable of withstanding the tests specified in 4.4.1.1, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.1.2 Vibration; ready nonoperating or ready condition. The firing switch, less pack or packaging and rigidly self-mounted on a vibration device as specified in 4.4.1.2, shall be capable of withstanding vibrations of 3 g's over the frequency range of 10 to 100 Hertz (Hz) in each of three mutually perpendicular planes for time periods and temperatures specified in 4.4.1.2. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.4.1.3 Vibration; operating condition. The firing switch, self-mounted on a vibration device as specified in 4.4.1.3, shall be capable of withstanding vibrations of 1 g over the frequency range of 20 to 150 Hz in each of three mutually perpendicular directions for time periods and temperatures specified in 4.4.1.3. Thereafter, the switch shall shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

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3.4.2 Shock

3.4.2.1 Shock; transportation, storage, and handling. The firing switch shall be capable of withstanding the test specified in 4.4.2.1, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2.2 Shock; ready condition. The firing switch, self-mounted on a shock testing device as specified in 4.4.2.2, shall be capable of withstandinging a shock of 100 g peak magnitude, of not less than 8 ms pulse duration, and approximating a one-half sine wave pulse. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.4.3 Acceleration. The firing switch, mounted securely on an acceleration device as specified in 4.4.3, shall be capable of withstanding an acceleration of 15 g's applied for not less than 0.2 second. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.4.4 Temperature and humidity

3.4.4.1 Transportation and storage condition. The firing switch, packaged and packed as defined in Section 5, shall be capable of withstanding transportation and storage temperatures and humidities between minus 65 degrees F and plus 160 degrees F for a period of 28 days as specified in 4.4.4.1. Thereafter, the switch, less pack or packaging, shall show no corrosion or damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.4.4.2 High and low temperature operation. The firing switch, less pack and packaging, shall be capable of withstanding temperatures over the range of minus 40 degrees F to plus 130 degrees F, as specified in 4.4.4.2 and 4.4.4.3. At the conclusion of each test, and while still at the test temperature, the switch shall show no corrosion or damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. The firing switch, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to

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neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, staking, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the firing switches shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five firing switches, manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspections of 4.5. These samples shall be subjected to the tests of 4.4 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from three consecutive groups, shift to Step 2.

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Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from three consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration tests

4.4.1.1 Vibration; transportation, storage, and handling. The firing switch, packaged and packed as specified in Section 5, shall be subjected to the appropriate vibration testing specified in OR-11.

4.4.1.2 Vibration; ready nonoperating or ready condition. The firing switch, mounted as shown on Dwg 1576934, shall be subjected to the vibration testi of Schedule I in each of three mutually perpendicular axes. Thereafter, the switch shall show no damage and shall meet all performance requirements and product characteristics of this specification.

4.4.1.3 Vibration; operating condition. The firing switch, mounted as shown on Dwg 1576934, shall be subjected to the vibration scheduling of Schedule II in each of three mutually perpendicular axes. During vibration testing, the firing switch shall be continuously monitored to ensure that it shows no evidence of closure. At the conclusion of vibration testing, the firing switch shall meet all the performance requirements and product characteristics of this specification.

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Schedule I

Frequency Range	Amplitude	Minimum Test Duration Room Temp.	-20°F	+130°F
10 - 30 Hz	0.060 ± 0.006 in.	60 min.	15 min.	15 min.
31 - 60 Hz	3g ± 0.3g	60 min.	15 min.	15 min.
61 - 100 Hz	3g ± 0.3g	60 min.	15 min.	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a 3 to 5 minute period, for the duration of the test.

As an alternate method, the firing switch shall be vibrated at specific frequencies, differing by not more than 5 Hz in each of the test axes and covering the entire frequency range and amplitudes of Schedule I. The total test time shall be not less than that of Schedule I and the test times at the specific frequencies shall be equally divided.

Schedule II

Frequency Range	Amplitude	Minimum Test Duration Room Temperatures
20 - 60 Hz	1g ± 0.1g	15 min.
61 - 150 Hz	1g ± 0.1g	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a three to five minute period, for the duration of the test.

As an alternate method, the firing switch shall be vibrated at specific frequencies, differing by not more than 5 Hz in each of the test axes and covering the entire frequency range and amplitude of Schedule II. The total test time shall be not less than that of Schedule II and the test times at the specific frequencies shall be equally divided.

4.4.2 Shock tests

4.4.2.1 Shock; transportation, storage, and handling. The firing switch, packaged and packed as specified in Section 5, shall be subjected to the appropriate shock testing specified in OR-11.

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4.4.2.2 Shock; ready condition. The firing switch, mounted as shown on Dwg 1576934, shall be subjected to three shock pulses of not less than 8 ms duration and approximately one-half a sine wave. The pulses shall have a magnitude of 150 plus or minus 15 g's and shall be applied in both directions along each of three mutually perpendicular axes (total of 18 shock pulses). Thereafter, the firing switch shall show no damage and shall meet all performance requirements and product characteristics of this specification.

4.4.3 Acceleration. The firing switch, mounted as shown on Dwg 1576934, shall withstand 15 plus or minus 2 g's acceleration for a period of not less than 0.2 second in each of three mutually perpendicular axes with inertial loads in both directions. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

4.4.4 Temperature and humidity tests

4.4.4.1 Transportation and storage temperature. The firing switch shall be subjected to temperature and humidity cycling as specified for a nonoperating torpedo in Section 4 of MIL-T-18404.

4.4.4.2 High temperature operation. The firing switch, less pack and packaging, shall be placed within a test chamber and the temperature shall be raised to plus 130 degrees F plus or minus 5 degrees F and maintained for a period of 48 plus or minus 5 hours. At the end of this period and while still at the test temperature, the switch shall meet all the performance requirements and product characteristics of this specification.

4.4.4.3 Low temperature operation. The firing switch, less pack and packaging, shall be placed in a test chamber and the temperature shall be lowered to minus 40 plus or minus 5 degrees F and maintained for a period of 48 plus or minus 5 hours. At the end of this period and while still at the test temperature, the switch shall meet all the performance requirements and product characteristics of this specification.

4.4.5 Operating life

4.4.5.1 Definition of one cycle. One cycle of operation for the firing switch consists of the discharge of a 4.0 microfarad (mfd) capacitor, having an initial charge of 150 volts dc, through a 5 ohm resistor with one impact applied to the switch.

4.4.5.2 Operating life test. Mount the firing switch in a test fixture that will impart sufficient shock to trip the switch. Connect the circuit (4.4.5.1) for discharging the 4.0 mfd capacitor upon switch closure. Subject

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the firing switch to 100 cycles of operation, rotating the switch approximately 3-1/2 degrees with respect to the line of impact prior to each successive cycle. The minimum time between each impact shall be one minute.

4.4.6 Functional tests. At the conclusion of tests specified in 4.4.1.1 and 4.4.2.1, functional and other tests, as applicable, shall be conducted on the firing switch to determine freedom from operational malfunction and compliance with all requirements of Section 3.

4.4.6.1 Waiver of tests. Tests specified in 4.4.1.1 and 4.4.2.1 may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.5 Quality conformance inspection. Each Firing Switch Mk 9 Mod 0 delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirements specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, Firing Switch Mk 9 Mod 0 shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature: Room ambient plus 65 degrees Fahrenheit (F) to plus 95 degrees F (plus 18 degrees centigrade (C) to plus 35 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, where definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 An evacuation chamber that allows observation of its interior, together with a vacuum pump and suitable gauge capable of registering reduced pressures in the order of 15 inches of mercury.

4.5.3.2 A centrifuge disposed to rotate in a horizontal plane and capable of imparting acceleration up to 10 gravity units. Provision shall be made

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for mounting the firing switch in a socket with its axis parallel to the plane of rotation, with additional provision for rotating the switch in the plane of rotation to an angle of 180 ± 1 degrees, when viewing the table from topside. Provision shall be made for mounting the firing switch in a socket with its axis vertical and its base down, with additional provision for rotating the switch about its axis in 45 ± 1 degree increments. Slip-rings and brushes, or some other appropriate arrangement, shall be provided for making electrical contact with the rotating switch.

4.5.3.3 An oscilloscope whose ripple is limited to less than 0.1 percent. The oscilloscope shall have associated with it an electrical circuit containing a 24 volt dc power source, a 250 ohm resistor, a 1.0 ohm precision resistor, a 0.5 ohm resistor, a single-pole three-position calibration switch, a single-pole single-throw switch, and the firing switch socket on the centrifuge. The dc potential across the 1.0 ohm resistor is applied to the vertical deflection plates of the scope.

4.5.3.4 A megohmmeter with an integral source of 500 volts dc and a mid-scale reading in the order of 10 megohms.

4.5.3.5 A standard electrical power source of 110 volts ac, 60 Hz, for operation of the vacuum pump, centrifuge motor, oscilloscope, and like equipment.

4.5.4 Test procedure. The quality conformance tests shall be performed in accordance with the following procedures:

4.5.4.1 Airtightness test. This test shall be performed upon the firing switch inner assembly, Dwg 771396, before it is finally assembled into the plastic case. Fill a container sufficiently deep with water so that the firing switch inner assembly can be completely submerged. Place this water bath in the evacuation chamber and reduce the air pressure to at least 15 inches of mercury in order to remove the air which is dissolved in the bath. Bring the pressure back to ambient and submerge the firing switch inner assembly in the deaerated water bath. Evacuate the chamber to a pressure of 15 inches of mercury (maximum), and observe the switch for a period of 2 minutes. Any leakage from the switch inner assembly, as evidenced by escaping air bubbles, shall render the switch unacceptable.

4.5.4.2 Sensitivity and contact resistance test

4.5.4.2.1 Calibration of oscilloscope. Before performing the sensitivity and contact resistance test, it is first necessary to calibrate the oscilloscope to represent a one ohm deflection in fractions thereof. Connect the test equipment as shown on the wiring diagram (Figure 1). The resistor cut-out switch shall be in the open position for the entire calibration. Throw the calibration switch to position 1 and adjust the sweep so it is deflected

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five scale divisions below center scale. Then throw the calibration switch to position 3 and adjust the sweep deflection five scale divisions above center scale. Repeat these steps until the sweep remains properly positioned when the calibrate switch is changed from position 1 to position 3 and the centrifuge is operating. The scope is now calibrated so that each scale division represents one-tenth of an ohm, that is, the relationship between scale deflection and resistance will be linear.

4.5.4.2.2 Conversion of revolutions per minute (rpm) to gravity units. Draw a table or graph of rpm versus gravity units according to the formula

$$A = 3.4 \times 10^{-4} RN^2$$

where the following units apply:

A = gravity units of acceleration

R = radial arm of centrifuge in feet, measured from center of rotation to center of ball in firing switch

N = revolutions per minute

If the centrifuge does not register rpm directly, but indicates angular velocity in some other arbitrary unit, it will be necessary to calibrate the centrifuge in terms of rpm and adjust the graph accordingly.

4.5.4.2.3 Test. Mount the firing switch in the socket on the centrifuge (4.5.3.2) so that its axis is perpendicular to the plane of rotation. Throw the calibration switch into position 2, close the resistor cutout switch, and start the centrifuge. After attaining an acceleration of 2 gravity units, increase the acceleration thereafter at a rate not exceeding 0.1 gravity units per second. When a straight line dc presentation is observed on the oscilloscope screen, note the reading on the centrifuge indicator. Convert this reading to gravity units in accordance with 4.5.4.2.2, and record this as the value for steady closure. Open the resistor cutout switch and record the sweep deflection in ohms resistance for the firing switch during steady closure. If the particular circuit warrants it, short out the firing switch as close to the socket as is physically possible, and obtain a scope deflection for the circuit resistance alone, while the centrifuge is still at the same speed. This resistance shall be subtracted from the value previously obtained to determine true switch contact resistance in the position tested. When mounted so that the vertical switch axis is perpendicular to the plane of rotation and the switch is in a base-down position, test the firing switch for instantaneous and steady closure contact resistance in seven other positions at 45 ± 1 degree increments of rotation about its axis. Then mount the switch so that its axis is parallel to the plane of rotation. Test

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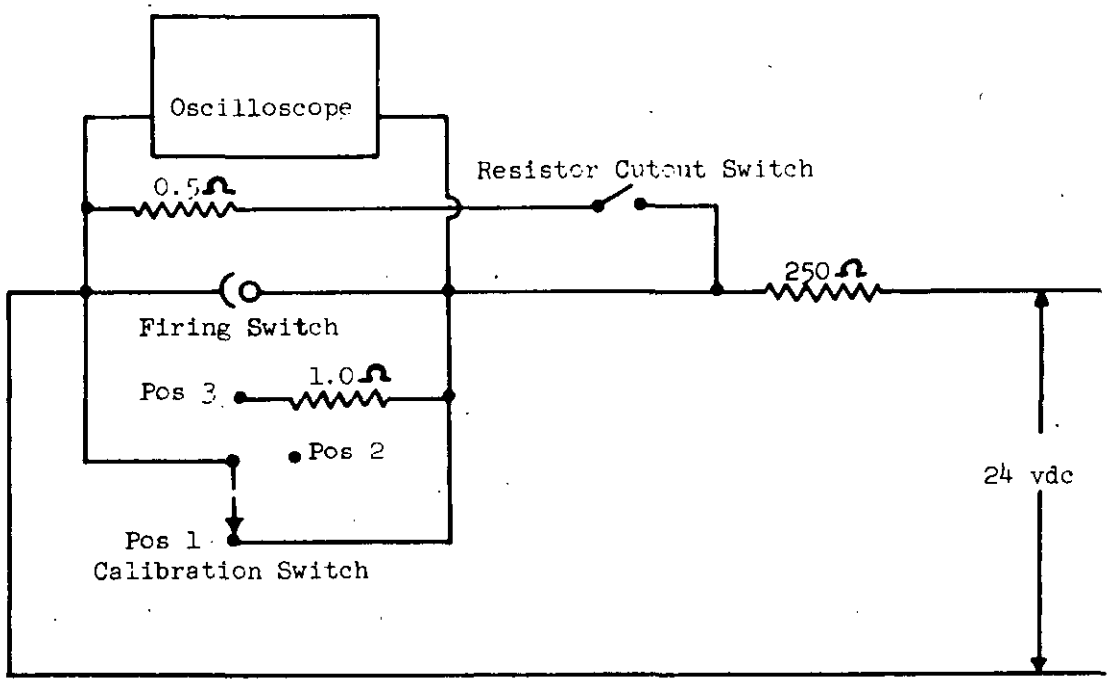


Figure 1. Sensitivity and Contact Resistance Test Wiring Diagram

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the firing switch for instantaneous and steady closure acceleration and steady closure contact resistance in two positions such that the ball is accelerated towards the nose cap and then towards the base.

4.5.4.3 Dielectric test. Apply the test probes of the megohmmeter to the terminals of the firing switch when it is in the normally open condition. The insulation resistance shall be not less than 10 megohms.

4.5.4.4 Visual and dimensional examination. Each firing switch shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.5 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified in the contract or order (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the firing switches shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the firing switches shall be dried in accordance with one or more of the procedures given in MIL-P-116, provided the procedure is not injurious to the item.

5.1.1.3 Preservative application. Not applicable.

5.1.1.4 Unit packaging. Unit packaging of the firing switch shall be in accordance with Method IA-15X of MIL-P-116. The unit package shall contain one firing switch.

5.1.1.5 Intermediate package. Twenty firing switches, packaged in accordance with 5.1.1.4, shall be placed in a close fitting intermediate container conforming to PPP-B-665.

5.1.1.6 Cushioning. The unit package, intermediate package, and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the firing switch will meet all the requirements of Section 3 after being tested in accordance with 4.4.1.1 and 4.4.2.1. At the time of packaging and packing, the moisture content of the cushioning material shall not exceed 5 percent. The pH range of the material shall be between 6.0 and 8.0.

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5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning and drying. Cleaning and drying shall be in accordance with 5.1.1.

5.1.3.2 Preservative application. Not applicable.

5.1.3.3 Unit packaging. Unit packaging shall be in accordance with Method III of MIL-P-116. The unit package shall contain one firing switch.

5.1.3.4 Intermediate package. Twenty firing switches, packaged as specified in 5.1.3.3, shall be placed in a close fitting intermediate container conforming to PPP-B-665.

5.2 Packing. Packing shall be Level A, B, or C, as specified in the contract or order (see 6.2).

5.2.1 Level A

5.2.1.1 Exterior containers. Two hundred firing switches packaged as described in 5.1.1 shall be packed in a shipping container conforming to Type I or Type II, Class 3, of PPP-B-636.

5.2.2 Level B

5.2.2.1 Exterior containers. Two hundred firing switches packaged as described in 5.1.1 shall be packed in a shipping container conforming to Type I or Type II, Class 1, of PPP-B-636.

5.2.3 Level C. Firing switches packaged in accordance with 5.1.3.4 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to the markings required by contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. Firing Switch Mk 9 Mod 0 is intended for use in Exploder Mechanism Mk 19 Series.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exception to this specification.

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Assigned activity for preproduction and periodic production inspections.

(f) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID
4.4	Preproduction and periodic production inspection	

DIDS (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N297

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-22135C(OS) Switch, Firing, Mark 9 Mod 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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